

BEVISONNEERS

THE MERCEDES-BENZ FELLOWSHIP

Planet-positive Project Portfolio
BioViL

PANKAJ SHARMA

Vision Statement

Our Vision is to build just communities

by providing them access to clean, alternative energy sources (currently bio-energy)

through education and innovative technological interventions

to reduce health risks, promote equity, and foster a sense of belonging,

ultimately creating self-sustaining, and resilient communities.



Design Plan

My project addresses challenges within **Energy Decarbonisation and Universal Access** and focuses on **implementing** and **optimizing biogas production** from **animal, organic, and household waste** by leveraging **community engagement** (e.g., conducting training workshops, youth programs, etc), **technology** (e.g., developing user-friendly monitoring tools for feedstock management), and **iterative prototyping** (e.g., creating and refining simple tools based on farmer feedback)



Individual and
Collective action



Science and
technology



Capacity building

"How-might-we" Questions

- How might we increase awareness of biomass resource potential in rural Indian communities?
- How might we foster a sense of ownership among community members in the bioenergy projects?

- How might we retrofit the primitive solutions to enhance the system performance and overall efficacy?
- How might we assess the suitability of existing bioenergy technologies?

- How might we ensure long-term knowledge retention and knowledge transfer within the community?
- How might we address the shortage of expertise for maintaining bioenergy installations?

ENTRY POINT: Energy Decarbonisation and Universal Access

Tools and Principles

- **"Inclusive Energy Circles"**: Workshops and meetings designed to involve communities in bioenergy discussions
- **"Bioenergy Simulation Game"**: A role-playing tool that allows community members to experience different aspects of managing a bioenergy project.

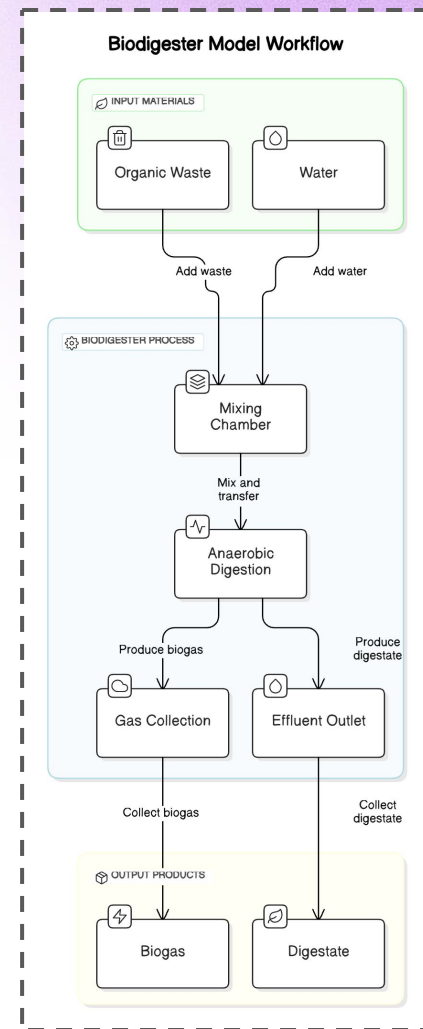
- **"Adaptive Technology"**: Research led technological modifications to the existing low-cost solutions for increased performance and output fitting to rural settings

- **"Knowledge Retention and Transfer Program"**: Digital Platform for documenting and sharing the knowledge within communities
- **"Integrational Bioenergy Mentorship Program"**: Initiative pairing experienced practitioners with youth to enhance engagement and develop necessary skills

Prototyping Envisioning

Visually depicting the **biodigester system** for the first time provided clarity on the entire process flow and highlighted the interactions between each component, emphasizing the critical role of the **digestion cycle**, and **proper feedstock management** in energy production efficiency. This representation revealed potential bottlenecks, such as gas storage capacity and energy output management, which are essential for optimizing system performance.

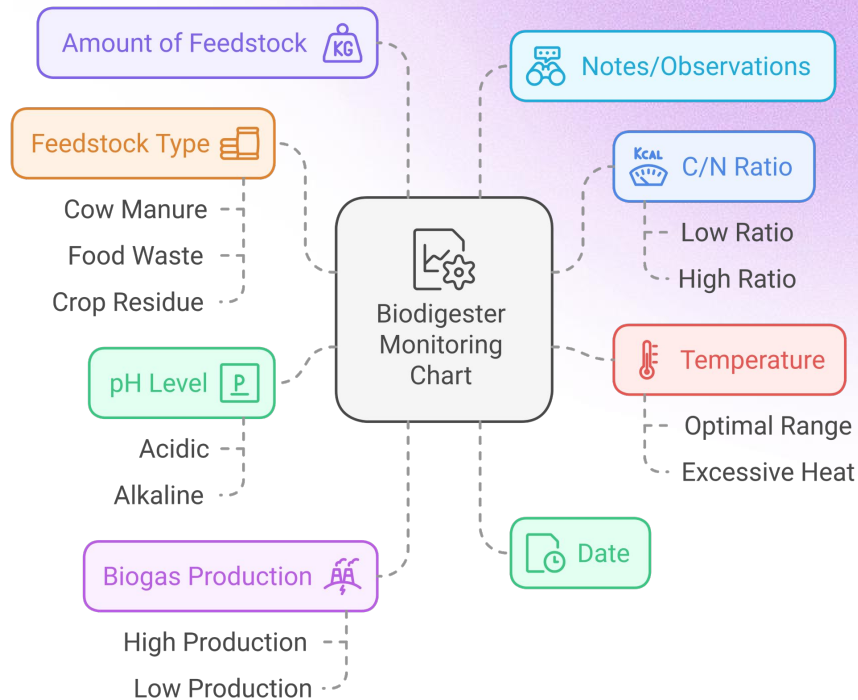
Additionally, I learned that selecting the right tools is vital for effectively conveying the system's complexity and engaging community members. Tools like community mapping workshops and participatory decision-making platforms not only enhance understanding but also foster local ownership of the project, ensuring that our approach remains accessible and relevant to all stakeholders.



Prototyping Chronicles: *Basic Monitoring Chart for Biodigester Management*

The **Basic Monitoring Chart** prototype aimed to help farmers manually track **feedstock inputs**, **biogas production**, and other key parameters over time. However, the implementation faced several challenges, including **inconsistent usage** due to the chart's complexity, **lack of real-time feedback**, **difficulties in data interpretation**, and **limited engagement** from users.

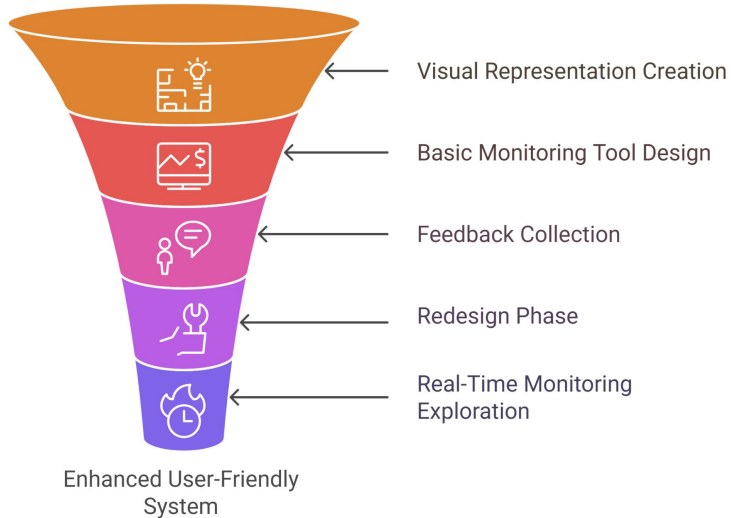
These issues highlighted the need for a simpler design that encourages **regular use**, **real-time monitoring capabilities** for immediate feedback, and **comprehensive training** to help farmers understand how to analyze the data effectively. In response, a refinement process was initiated to redesign the chart for better usability and to explore automated monitoring solutions while developing educational programs on best practices for biodigester management.



Minimum Viable Solution

Biodigester Prototype Evolution

Initial Concept Development



In summary

I got to prototype a project that addresses challenges within **Energy Decarbonization and Universal Access** and focuses on **implementing and optimizing biogas production from animal, organic, and household waste** by leveraging **community engagement** (e.g., conducting training workshops and youth programs), **technology** (e.g., developing user-friendly monitoring tools for feedstock management), and **iterative prototyping** (e.g., creating and refining simple tools based on farmer feedback) to test solutions.

The prototyping approach involved creating a **Basic Monitoring Chart** to help farmers manually track key parameters such as feedstock inputs and biogas production. While this tool provided valuable insights into the biodigester system, it faced challenges like complexity, inconsistent usage, and a lack of real-time feedback. These issues highlighted the need for a more intuitive design and comprehensive training resources.

And I learned that selecting the right tools is crucial for effectively engaging community members and ensuring project relevance. Feedback from my coach and **users** emphasized the importance of simplifying designs and enhancing user engagement. My prototyping journey revealed insights that translated into actionable solutions, such as redesigning the monitoring chart for better usability and exploring automated data collection methods. This iterative process not only improved our technical approach but also fostered deeper community involvement, moving us closer to our vision of building resilient communities through accessible bio-energy solutions.